S(14): A COMPUTER-ASSISTED DETERMINATION

Theorem. S(14) = {2,3,5,7,11,13,17,19,23,29,31,37,41,43}
= Primes(43).
The elliptic curve is arbitrary, and the number field has exact degree 14.
The dyadic argument gives the global bound p <= 16641.

All arithmetic computations used below are exact. The certificate packet contains their inputs, generation scripts, and checking scripts. The final coverage check passes for all 1924 primes up to 16641. In particular, the proof does not rely on extending the published strange-character table beyond its stated range.

1. OCCURRENCE AND THE DEGREE CONDITION

DKSS proves S(7)=Primes(23). A quadratic extension of a septic field preserves its torsion and has exact degree 14. This supplies all primes through 23.

For each of 29,31,37,41,43 the file occurrences.json records an irreducible polynomial f of degree 14 and rational functions x=a,y=g(a)/h(a), where f(a)=0. Set

$$r=(x^2*y-x*y+y-1)/(x^2*y-x),\quad s=(x*y-y+1)/(x*y),$$
$$b=r*s*(r-1),\quad c=s*(r-1).$$

Then E: Y^2+(1-c)XY-bY=X^3-bX^2, P=(0,0), is an exact occurrence: verify_occurrences.m proves irreducibility, nonsingularity, P nonzero, and p*P=0 for every indicated prime p.

The degree-14 places at 37,41,43 come from van Hoeij’s explicit tables. For example the x-coordinate at 41 is a root of

$$x^{14}-8x^{13}+21x^{12}-11x^{11}-43x^{10}+61x^9+24x^8-80x^7$$
$$+x^6+56x^5-4x^4-23x^3+5x+1.$$

All coordinates, including y, are retained in the certificate packet. The examples at 29 and 31 were found by explicit specializations of the Sutherland--van Hoeij models. Their exact verification does not rely on Hilbert irreducibility or on existence in a different degree.

Conversely, a prime dividing E(K)_tors occurs as the order of a K-rational point. The resulting noncuspidal point on X_1(p) has degree dividing 14. For p>43, the possibilities 1,2,7 are excluded by the known classifications. It therefore has degree 14. Write D for its irreducible rational trace divisor. This is precisely the S_new(14) condition in Derickx--Stoll.

2. THE DYADIC GLOBAL BOUND

We use the previously proved dyadic doubling argument, with an explicit coefficient certificate replacing the coarse polynomial estimate. Let J=J_0(p), T its integral Hecke algebra, e the winding element, I=Ann_T(e), and A=J/IJ the optimal winding quotient. The group A(Q) is finite. Put

$$M=H^0(X_0(p)_{Z_2},\Omega^1)[I],\quad L=\pi^*\mathrm{Cot}(A_{Z_2}).$$

The integral coefficient pairing identifies M with the Z_2-dual of (T/I) tensor Z_2. Raynaud’s theorem gives

$$2M \subset L \subset M.$$

No equality L=M is assumed. The earlier dyadic theorem, reproduced in the packet, gives the precise lattice argument and references.

Take m=28 and M0=157. Put

$$T'_n=\sum_{a|n} \mu(n/a)T_a,\quad L_n=T'_{2n}-2*T'_n.$$

The integral spans of L_1,...,L_28 and
(T_2-3)T_1,...,(T_2-3)T_28 coincide. Also (T_2-3)Te is contained
in integral absolute homology. DKSS’s intersection formula, for
p>2*m*M0=8792 and p*u=1 modulo M0, is
 L_n e . {0,a/M0}=epsilon_M0(na)-epsilon_M0(nu/a), 1<=n<=28,
where epsilon is the indicator of the upper half of the nonzero residues.

For each u=1,...,156 this 28 by 156 matrix has rank 28 modulo 2.
The rows are retained in dyadic_jets.json. verify_auxiliary.py rebuilds
every entry and independently checks all ranks in Sage. A dependence
among T_1e,...,T_28e modulo 2Te, multiplied by T_2-3, would contradict
these integral intersection matrices. Thus the first 28 coefficient
map on M is surjective. The cotangent image of
 f_28:X_0(p)^(28) -> A, Z |-> pi[Z-28*infinity]
at 28*infinity contains 2*Z_2^28.

We recall the two local facts that account for the characteristic-2
issue. If a map has cotangent image containing 2*Z_2^m, its integral
formal pullbacks can be chosen as 2*x_i+Q_i(x), with Q_i quadratic
or higher. On 4*Z_2^m a nonzero vector of minimum valuation k>=2
cannot map to zero: one linear term has valuation k+1<2k.
If a degree-d divisor reduces to d*c modulo 2, write its elementary
symmetric polynomial as E_D(t)=1+2B(t). Then
 E_(2D)(t)=E_D(t)^2=1+4(B(t)+B(t)^2),
so its double agrees with 2d*c modulo 4, even for ramified or repeated
points. Finally the torsion reduction kernel of an abelian scheme over
Z_2 is killed by 2, since doubling sends it into the torsion-free formal
neighbourhood modulo 4.

Suppose p>(2^(14/2)+1)^2=16641. At every place above 2, Hasse’s bound
excludes good reduction. Additive reduction cannot support this prime
torsion; nonsplit multiplicative reduction has odd torsion bounded by
2^f+1. In split multiplicative reduction the point must be in the
component group, since p>2^f-1. Thus, after Atkin--Lehner on X_0(p),
the trace divisor D0 of the image reduces to 14*infinity. Its winding
image f_14(D0) is rational torsion reducing to zero, and hence
 f_28(2D0)=2*f_14(D0)=0.
The doubled divisor agrees with 28*infinity modulo 4. The coefficient
and local arguments just given force 2D0=28*infinity, a contradiction.
Since 8792<16641, we obtain p<=16641. Using 28 coefficients, rather
than 14, is essential to the correction for the cotangent defect at 2.

3. PUBLISHED EXCLUSIONS AND THE PRIME 127

We use Derickx--Stoll, arXiv:2505.14109v1, Table 2 and Corollary 7.5,
only for p<100000. For 151<=p<=16641, its small-prime alternative
fails because p^2>(2^19/325)*14+1. Its two CM alternatives give
p=2*14+1=29 or p=3*14+1=43. The remaining alternative requires
(p-1)/(2*str(p)) to divide 14. No prime in this interval satisfies it.

For p=131,137,139,149, Proposition 4.1 applies on X_1(p), with
<2>, ell=2,n=5. The gonality bound 325*(p^2-1)/65536 exceeds 70;
Remark 4.4 gives the rational 2-torsion hypothesis. The relevant
diamond orders are 65,34,69,74. None of these orders, or any integral
half or third of one, divides 14. Thus an irreducible degree-14
noncuspidal divisor cannot be a sum of such orbits.

These published criteria exclude 1893 primes and leave
 47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113,127.

The prime 127 has an additional exclusion on an intermediate curve.

Let H=<2> in Delta=(Z/127Z)^*/{+1,-1}. It has order 7. Since 127 is non-strange, (h-1)J_1(127)(Q) is finite for every diamond h. The gonality bound on X_1(127) exceeds 79 and hence 5*14. Proposition 4.1 with ell=2 and a=2, using Remark 4.4, therefore makes D a sum of H-orbits. In characteristic zero, an order-7 diamond has no noncuspidal stabilizers: elliptic-curve automorphisms modulo sign have order at most 3. Hence D descends to a rational noncuspidal effective divisor D′ of degree 2 on Y=X_1(127)/H.

We prove gon_Q(Y)>=20 by counting points over F_4. The finite-field census gives six supersingular diamond orbits of degree-14 closed points on X_1(127), of sizes 9,9,3,9,9,9, in total 48. quotient127.m checks that <2> fixes every representative as a closed point and acts freely on its seven geometric translates. Commutativity of diamonds and Galois extends these checks to all 48 closed points. Their images on Y are distinct closed points of degree 2. Indeed, a free H-fibre has seven geometric points, so one degree-14 point accounts for the entire inverse image of a degree-2 point. Thus

 #Y(F_4) >= 2*48 = 96,
 gon_Q(Y) >= gon_(F_4)(Y) >= ceil(96/5)=20.

Good reduction at 2 holds for this prime-to-2 quotient modular curve.

The image of <3> in Delta/H has order 9. Its difference from 1 has finite rational image on J_Y: pullback to J_1(127) has finite kernel, commutes with diamonds, and the latter curve is non-strange. Apply Proposition 4.1 to Y with ell=3,a=3,n=7,d=2. No rational 2-torsion hypothesis is needed, and 7*2<20. It follows that D′ is a sum of <3>-orbits. On noncuspidal characteristic-zero points these have length 9 or 3, even allowing j=0 or 1728. No sum has degree 2. This excludes 127. No canonical model or local nonprincipality test at level 127 is used in this argument.

4. RATIONAL-CUSP REDUCTIONS AT THE OTHER SIXTEEN LEVELS

We use the endomorphism version of the DKSS formal immersion criterion. The maps are actual integral endomorphisms with finite rational image, composed with t_q=T_q-<q>-q for odd q in the mu-level convention. Consequently they kill the whole rational Jacobian group. Fricke transports the criterion to the geometric point-level rational cusps. This avoids any optimal-quotient cotangent saturation assumption at 2.

The selected intermediate diamond indices are

 47:23,53:13,59:29,61:15,67:11,71:5,73:12,79:39,
 83:41,89:11,97:8,101:5,103:17,107:53,109:54,113:56.

At 61,67,73 several maps are combined. At 107,109,113 one can use (<r>-1)t_3 directly, since these levels are non-strange. The explicit operators and characteristic polynomials occur in the FI exporters.

Every rational-cusp multiplicity tuple of total degree 14 passes. Normalize a cusp of largest multiplicity to index zero. Apart from the balanced case 7+7, all other multiplicities are at most 6. Thus we retain 14 coefficients at index zero and 6 at the other indices, and check every normalized prefix. Separately, for every other cusp we check rank 14 for the two length-7 prefixes. These balanced pairs also all pass.

For a small binary kernel we enumerate all its nonzero relations.

A relation with required prefix lengths m0,m1,... can occur in a
normalized degree-14 tuple only if
 max(m0,m1,...)+sum_(i>0)m_i <=14.
Larger kernels are checked by direct prefix enumeration, with an
independent dynamic-programming count. Compression reconstructs all
original rows exactly. Sage independently verifies the selected
full ranks, every balanced-pair rank, and the spanning property of
the relation bases. The selected successful files are identified in
formal_immersion.json. The earlier index-4 experiment at 97 fails
and is not used; the final certificate there has index 8.

5. ORDINARY AND MIXED NONCUSPIDAL REDUCTIONS

census_sources.m enumerates every j-invariant over F_(2^f), f<=14,
all twists, and all cyclic subgroups in the rational p-torsion,
together with all scalar level structures. It identifies closed points
and full diamond orbits by universal Tate coordinates. If E[p] has
rank 2 it enumerates all p+1 projective directions; at p=127,f=14
this is essential and explicitly checks all 128 directions.

At the sixteen remaining local levels the minimum ordinary degree
is at least 7. Two ordinary points can therefore occur only with
degrees 7+7, at 59,61,67,71,73. A single ordinary point otherwise
has a rational-cusp complement, except for the explicitly listed
supersingular mixture below.

In geometric point-level notation set A_3=T_3-1-3<3>. It kills
rational torsion and every rational cusp as an actual divisor.
Use finite-rational-image endomorphisms alpha as follows:
 alpha=1 at 47,59,71;
 alpha=sigma^6-1 at 61;
 alpha=(sigma-1)*Phi_12(sigma) at 97;
 alpha=(sigma-1)*(sigma^5+1) at 101;
 alpha=sigma-1 at the other levels.
Here sigma is a generator of the diamond group. The published rank-zero
results and Table 2 justify these choices, so alpha*A_3 kills J(Q).
Nonprincipality of alpha*A_3 of the nonrational reduction part therefore
excludes that reduction of D.

We use the full canonical models and integral-at-2 lattices already
verified in the lower-degree work. Degree-at-most-13 source/Hecke jets
are reused only after direct Tate-coordinate checks of their diamond
orbits against the new complete census. Each old orbit must be matched
exactly once. The canonical bases are identical; the matching assertions
and their logs are retained. New degree-14 sources and all degree-3
isogenies and weighted trace targets are computed afresh. Target degrees
sum to 4*deg(source). Eight-term canonical jets use the ordinary local
parameter a6, with the division-polynomial derivative checked nonzero.
Identical support is identified before integral multiplicities and
cancellations are formed.

For completeness the nonprincipality obstruction is elementary.
Write alpha*A_3(P)=U-V with disjoint effective U,V. Put R=H^0(K^s),
s=2 at p<=53 and s=1 otherwise; dim R>deg U is checked. Choose
nonzero F in R(-U). If div(g)=U-V, then S=F/g is a nonzero element
of R(-V), and H*S belongs to F*R for every H in R(-U). The stored
homogeneous condition matrix on S has full rank, so such S does not
exist. Every chosen F and H is retained. Multiplication tensors are
verified by q-products beyond the relevant Sturm bound; support

conditions use actual jets. Sage independently checks the final ranks.

There are 1019 ordinary diamond orbits. The matrices exclude 1018;
the one remaining orbit at 113, of degree 8 and size 7, is treated
by a stabilizer below. It is not asserted nonprincipal.

For the two-ordinary-point cases, choose source orbits A,B with sizes
a,b and normalize the first point to its orbit representative. If
A!=B, relative shifts 0,...,gcd(a,b)-1 suffice. If A=B, use shifts
0,...,floor(a/2), since the pair is unordered. The numbers of pair
orbits at 59,61,67,71,73 are respectively 59,16,17,26,19: in total
137. All their alpha*A_3 sums are nonprincipal, including repeated
points. All 137 matrices pass the independent Sage rank checks.

There is one supersingular/ordinary mixture: at 73 a supersingular
degree-6 point, an ordinary degree-7 point, and one rational cusp.
The endomorphism sigma^4-1 fixes each supersingular closed point.
Its A_3 test on the single ordinary degree-7 diamond orbit is
nonprincipal. This excludes every such mixed reduction.

6. STABILIZERS AND THE SMALL GONALITY CERTIFICATE AT 53

The remaining supersingular degrees and orbit sizes are
 53:(13,1); 73:(6,4); 109:(9,3),(12,3); 113:(7,4).
The only nonrational cusp degrees at most 14 are
 73:9,89:11,113:14.
The census proves that no other mixtures are possible. Repeated
supersingular supports have degree 6+6 at 73 or 7+7 at 113.

We recall the stabilizer argument, including its degree issue.
Suppose h fixes the nonrational part of red_2(D), and lift its
rational-cusp complement to a rational cusp divisor C. Put Z=D-C,
y=[hZ-Z] in J(Q). If h has order m and (h-1)J(Q) is finite, then
y is torsion even when deg Z is nonzero: the identity
 m*y=(h-1)[mZ-(1+h+...+h^(m-1))Z]
has a degree-zero class in brackets. Since red_2(y)=0, we have 2y=0.
If J(Q)[2]=0, or if h has odd order and acts trivially on J(Q)[2],
then y=0 (in the latter case the norm identity gives m*y=0).

If y=0 and gon_Q(X)>14+deg C, the principal divisor
 hD+C-D-hC
must vanish. If only 2y=0 is known, it suffices to have
 gon_Q(X)>2*(14+deg C).
The disjoint cuspidal and noncuspidal supports imply hD=D and hC=C.
The rational cusps form a free diamond orbit, so m divides deg C.
Except at 53, use gon_Q(X_1(p))>=325*(p^2-1)/65536.

At 53 we need gon_Q(X_1(53))>=16, which is stronger than this
analytic bound. Here is a fresh short certificate. Over F_2 there
are 26 rational cusps and no other places of degrees 1 through 6.
If a function had degree at most 15, one of its three F_2-rational
fibres would contain at least 9 distinct rational points. Move that
fibre to infinity. Its nonrational pole part would have degree at
most 6, and is therefore empty. All poles would be rational cusps.
Writing C_all for the sum of all 26 rational cusps, the pole divisor
is bounded by
C_all+E, where E is effective, supported on rational cusps, and has
degree at most 6; pad E to degree 6.

The canonical differential conditions for C_all have rank 26.
Modulo these conditions, gonality53.py tests all normalized cusp
prefixes of total excess degree 6. It checks 76256 tuples, independently
counted, and each has rank 6. Therefore the canonical conditions for
C_all+E have full rank 32, and Riemann--Roch gives
 l(C_all+E)=32+1-32=1.
This excludes the alleged function, including inseparable functions.
Thus gon_(F_2)>=16 and hence gon_Q>=16. The canonical evaluation
matrix and the complete prefix check are retained.

At 53, h=<4> has order 13 and fixes the supersingular degree-13
closed point. The independently verified exact integral homology
computation on JOne(53) has zero common kernel modulo 2 for A_3,A_5,
so J_1(53)(Q)[2]=0. The stabilizer argument gives y=0. Our new
bound 16>14+1 would force 13|1, a contradiction.

At 73 use h=<2>, of order 9. The audited computation on the actual
JOne(73) proves h acts trivially on rational 2-torsion. It fixes
both the degree-6 supersingular points and the degree-9 cusps.
The analytic gonality bound exceeds 26. The rational complements
have degrees 8,2,5 for one supersingular point, two such points,
and a degree-9 cusp respectively. None is divisible by 9.

At 89, h=<2> has order 11. The doubled-divisor bound is satisfied,
since gon>39>2*(14+3). A degree-11 cusp would require 11|3.

At 109, h=<23> has order 18 and fixes the degree-9 and degree-12
supersingular points. The gonality bound exceeds 58, and hence
2*(14+5) and 2*(14+2). The required 18|5 or 18|2 is impossible.

At 113, the exceptional ordinary degree-8 orbit is fixed by h=<40>,
of order 8. Since gon>63>2*(14+6), it would require 8|6.
For a single degree-7 supersingular point use h=<2>, of order 14,
which fixes the closed point. Again gon>63>2*(14+7), and 14|7 fails.
All new stabilizers are checked directly on Tate coordinates.

Finally, two degree-7 supersingular points, or one degree-14 cusp,
have C=0 and are fixed in reduction by h=<2>, of order 14. The
bound gon>63>28 forces hD=D. The geometric h-orbit length is 14,
or possibly 7 at j=1728. In the first case D is one orbit and maps
to a noncuspidal rational point on X_0(113), contrary to Mazur’s
prime-isogeny theorem. In the shortened case the p-subgroup is
stable under the CM automorphism. The underlying X_0(p) point
has field Q(i), so the standard CM degree bound used in the proof
of Derickx--Stoll Corollary 7.3 gives p-1<=2*[Q(D):Q]. But
112>28. Thus this case is impossible as well. There is no shortening
by 3, since 3 does not divide 14.

These cases exhaust all reductions at 2 and complete the sixteen
local exclusions. Together with Section 3 and the dyadic bound,
they prove the theorem.

7. VERIFICATION AND PRIMARY REFERENCES

final_coverage.json accounts for all 1924 primes up to 16641:
14 occurrence primes, 1893 published exclusions, the quotient-curve
exclusion at 127, and 16 local exclusions. There are 1018 ordinary,
137 pair, and one mixed-support full-rank obstruction matrices.
The selected formal-immersion ranks, balanced pairs, dyadic matrices,

and the extra gonality computation are independently checked as stated.

The earlier canonical models and rational 2-torsion inputs are reused from the audited lower-degree computations, with exact identification of bases and source orbits. The new degree-14 proof has the exact arithmetic checks described above; this does not assert a separate agent audit of the complete degree-14 argument.

Primary references:

- Derickx--Kamienny--Stein--Stoll, Torsion points on elliptic curves over number fields of small degree, Algebra Number Theory 17 (2023), 267--308; arXiv:1707.00364. The septic classification, integral intersection formula, and endomorphism formal-immersion criterion.
- Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109v1. Proposition 4.1, Remark 4.4, Corollary 7.5, Table 2, and the CM argument in the proof of Corollary 7.3. The table is used within its stated range.
- Mazur, Rational isogenies of prime degree, Invent. Math. 44 (1978), 129--162, for the nonexistence of a rational 113-isogeny.
- Van Hoeij, Low degree places on the modular curve X_1(N), arXiv:1202.4355, and the accompanying explicit degree-14 places.
- The earlier dyadic theorem in asymptotic_prime_bound.tex, reproduced with its Raynaud, Abbes--Ullmo, and Manin-constant references. The coarse coefficient estimate is replaced by the rank-28 certificate.